

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

497
500

ROTATIONAL MOTION

LECTURE-3

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Be Covered

- 1** Revision ✓
- 2** Angular Momentum ✓
- 3** Direction of Angular Momentum ✓
- 4** Torque ✓
- 5** Direction of Torque ✓

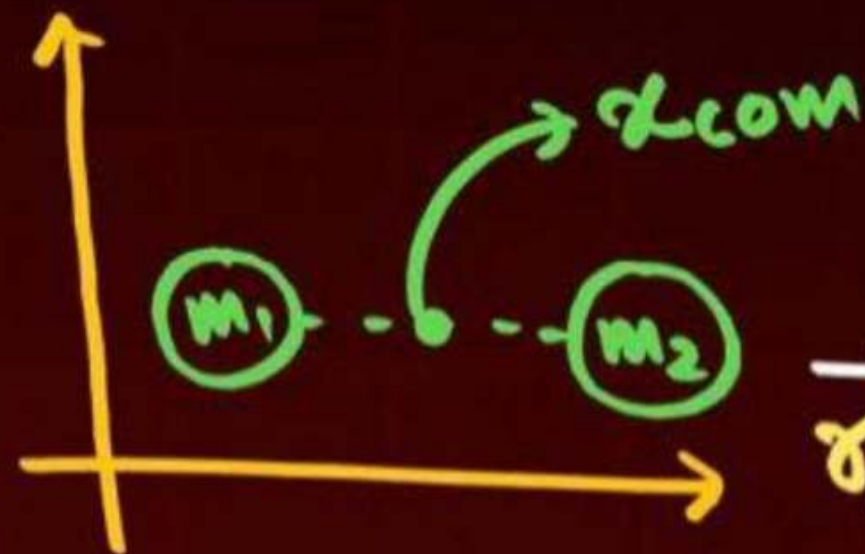
Recap From 1st Lecture

Position of Com

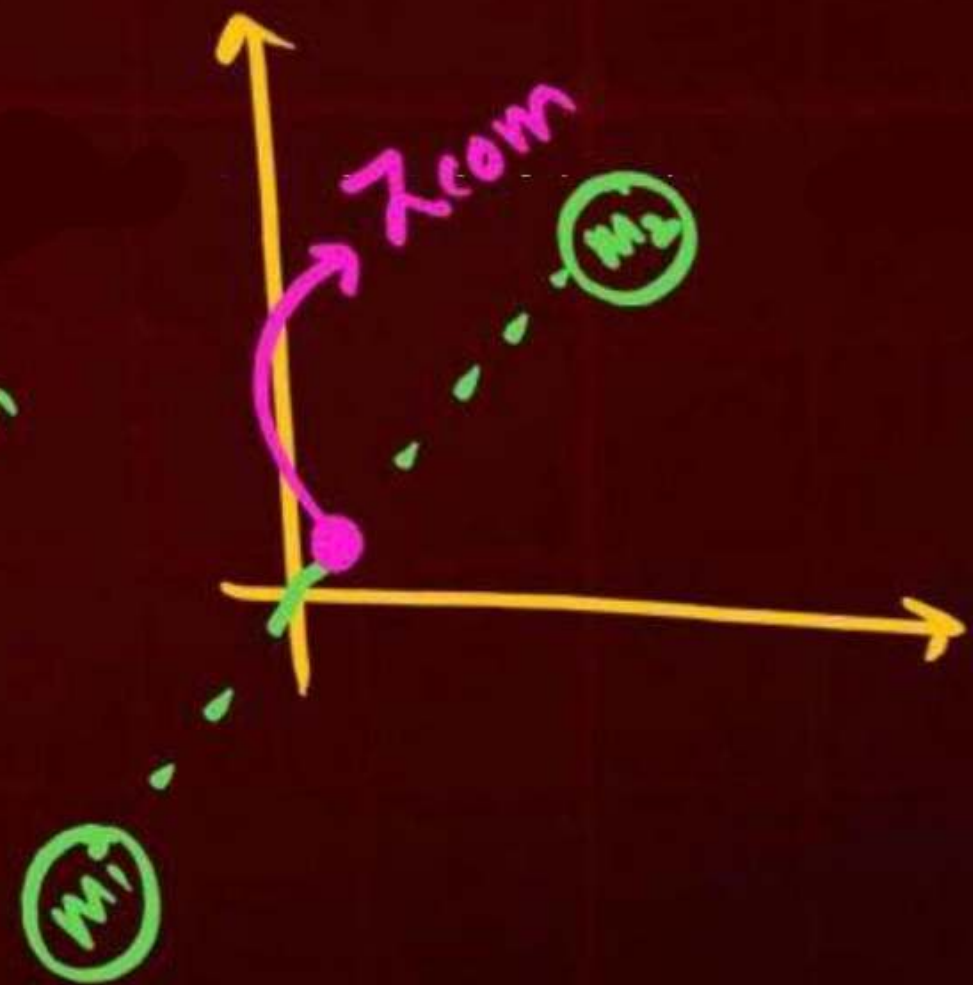
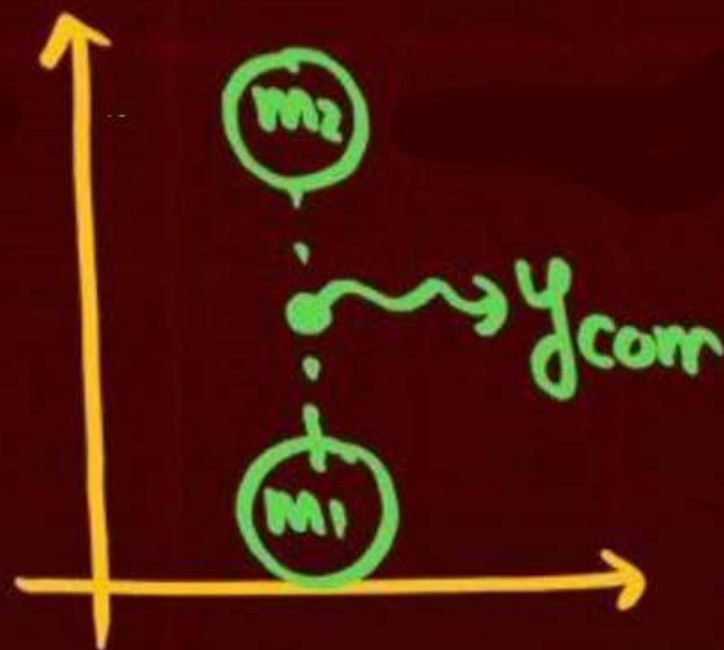
$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$y_{com} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

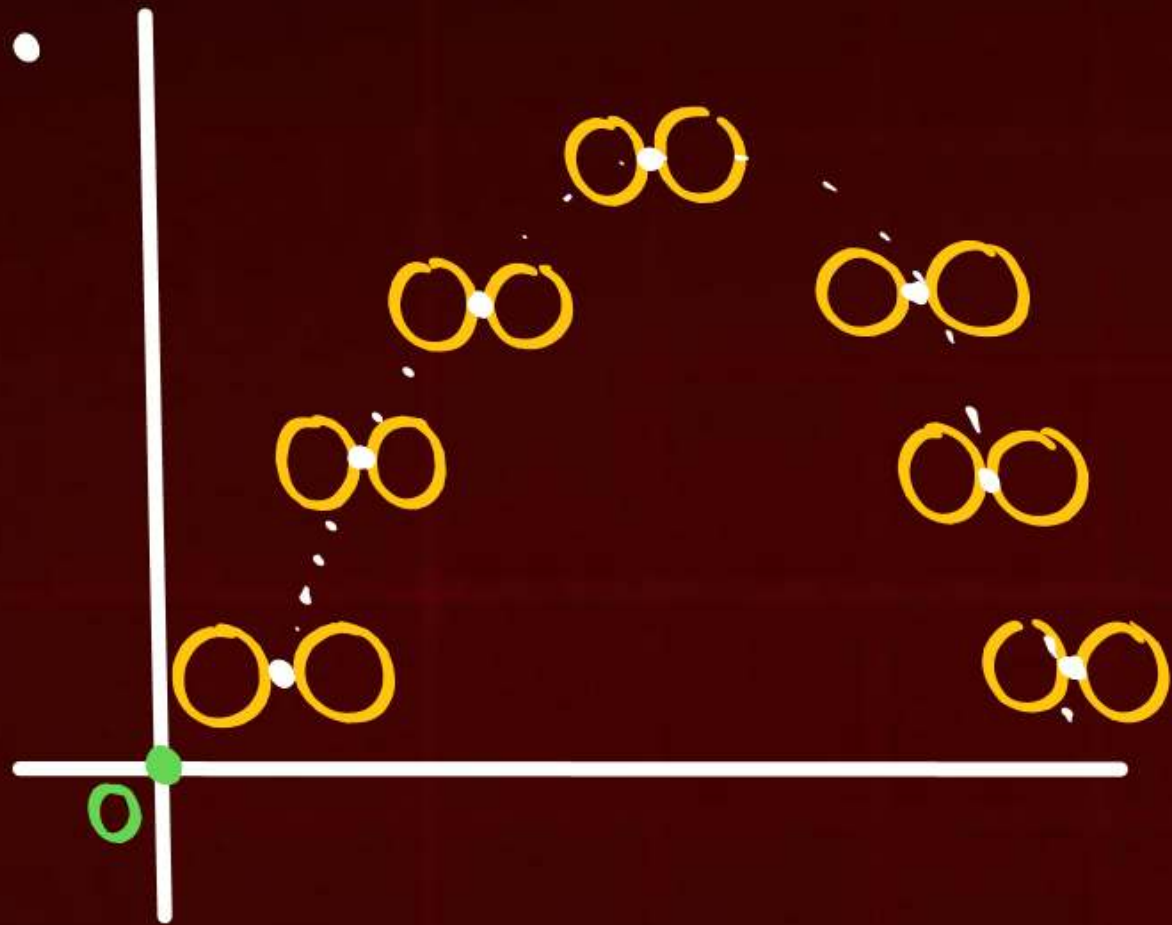
$$z_{com} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$



$$\vec{r}_{com} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$



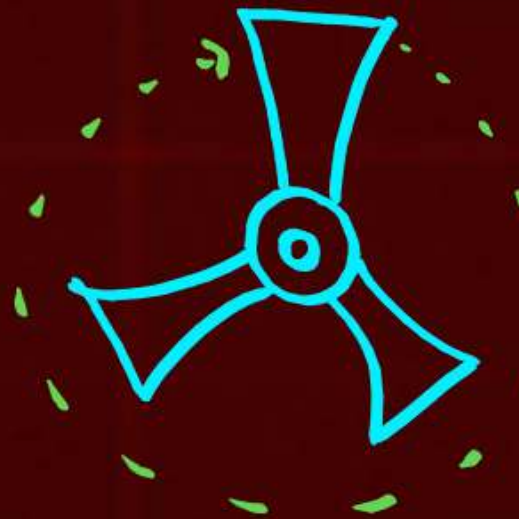
RECAP FROM 2nd LECTURE



$$\textcircled{1} V_{\text{com}} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$\textcircled{2} a_{\text{com}} = \frac{m_1 a_1 + m_2 a_2}{m_1 + m_2}$$

Rotational Motion



✓ Angular displacement

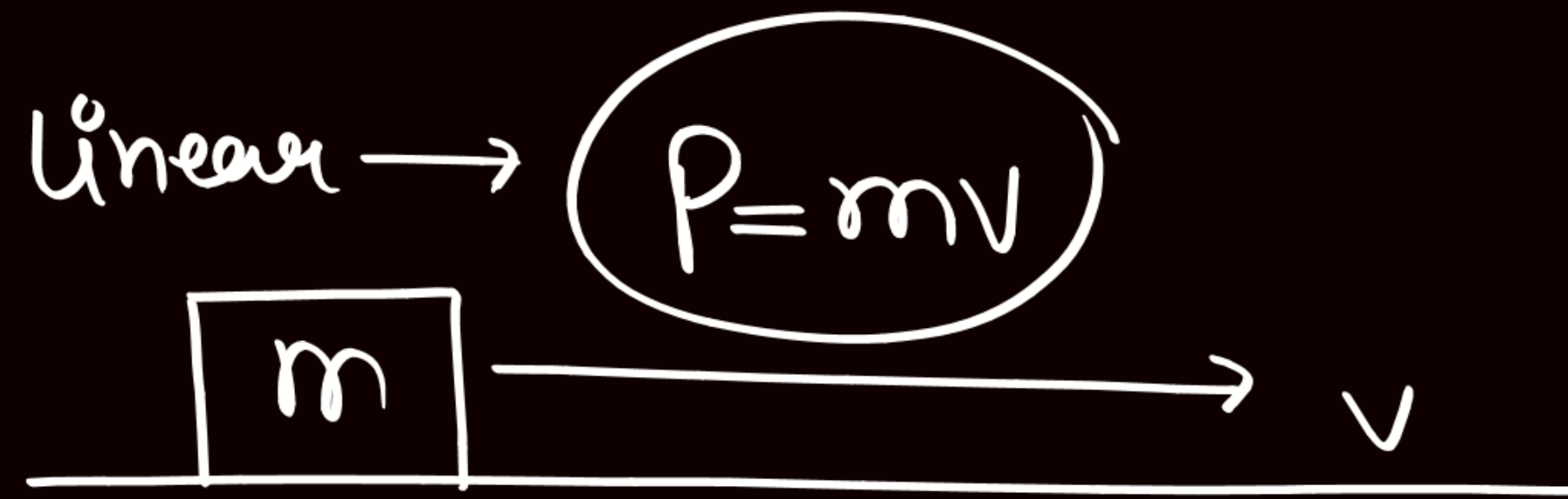
$$\theta = \frac{\text{Arc}}{R} \quad x = R\theta \quad \text{Rad}$$

✓ Angular velocity

$$\omega = \frac{\theta}{t} \quad v = R\omega \quad \text{Rad/sec}$$

✓ Angular acceleration

$$\alpha = \frac{\omega}{t} \quad a = R\alpha \quad \text{Rad/sec}^2$$



Angular Momentum

The momentum of the body when the body is rotated is called angular momentum

OR
Angular momentum ($\vec{L}$) is the cross product
of $\vec{r}$ and $\vec{p}$

$$\vec{L} = \vec{r} \times \vec{p}$$

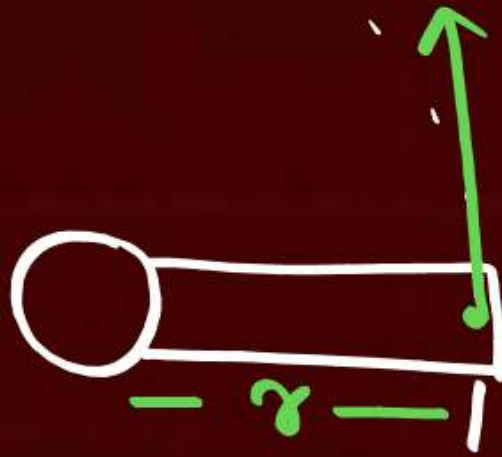
$$\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$$

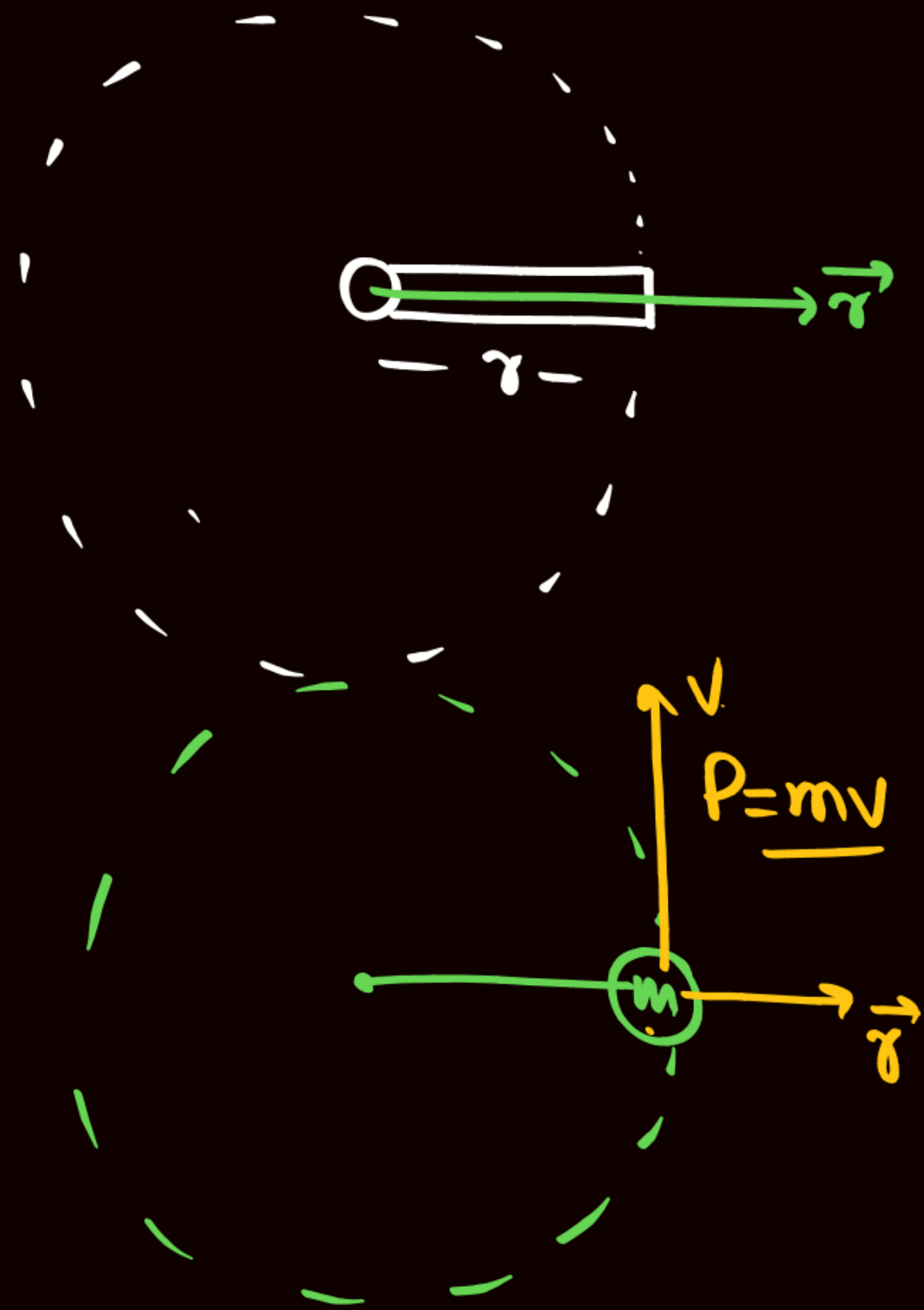
$$\vec{L} = \underbrace{r p \sin \theta}_m \hat{n}$$

Direction

↓
Screw Rule

$$\text{Unit} = \text{Kg} \frac{\text{m}^2}{\text{s}}$$





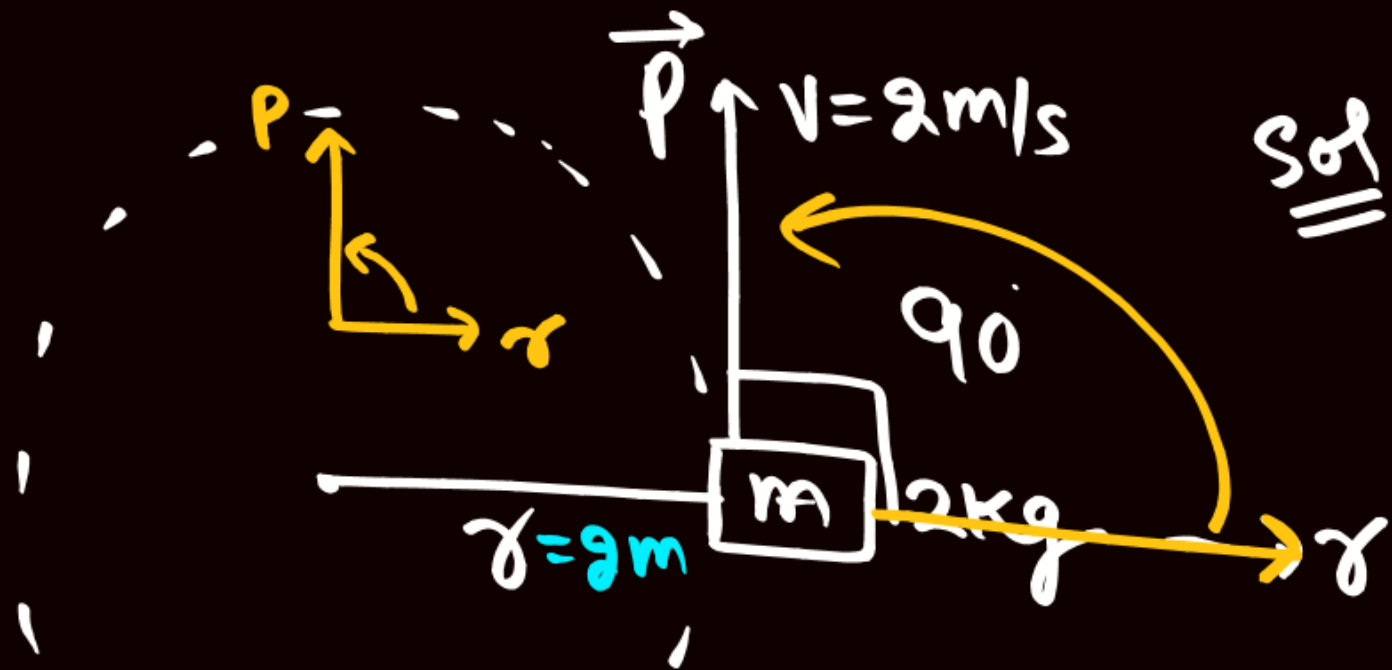
$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = r p \sin \theta \hat{n}$$

Direction of $\vec{L}$
 $\downarrow$
 Screen Rule

Q Numerical (Neet-2017)

Calculate the Angular momentum $\vec{L}$?



Sol

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = r p \sin \theta \hat{n}$$

$$\vec{L} = 2 \times 4 \sin 90^\circ \hat{n}$$

$$\vec{L} = 8 \times 1 \hat{n}$$

$$\vec{L} = 8 \hat{n}$$

Direction
Screw Rule

$$\vec{L} = 8 \text{ outward}$$

$$\vec{L} = \underline{8 \hat{k}} \text{ kg m}^2/\text{s}$$

$$\begin{aligned} p &= mv \\ &= 2 \times 2 \\ &= 4 \end{aligned}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

Q Numerical (Neet-2017)

Calculate the Angular momentum $\vec{L}$?

Sol

$$\vec{L} = \vec{r} \times \vec{p}$$

$$\vec{L} = r p \sin \theta \hat{n}$$

$$= 2 \times 4 \sin 90^\circ \hat{n}$$

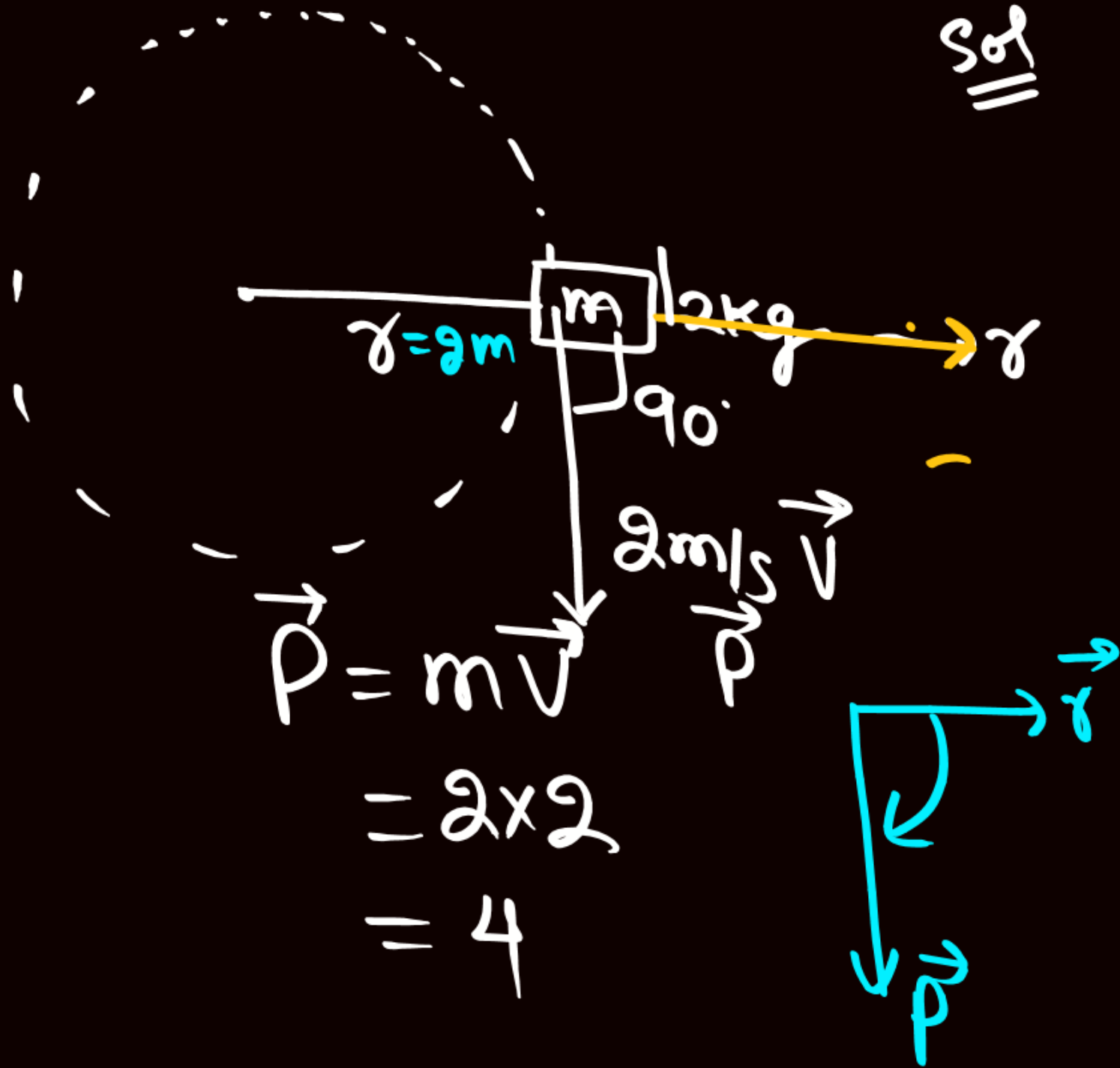
$$= 8 \times 1 \hat{n}$$

$$= 8 \hat{n}$$

$$= 8 \text{ inward}$$

$$= 8 \times -\hat{k}$$

$$= \underline{-8 \hat{k} \text{ kg m}^2/\text{s}}$$

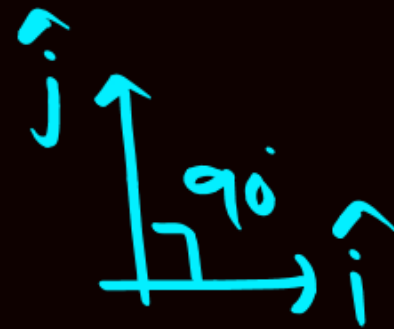


Q Next - 2019

$$\vec{r} = 2\hat{i} + 3\hat{j}$$

$$\vec{p} = 2\hat{i} + 5\hat{j}$$

$$\boxed{\vec{L} = ?}$$



(Numerical)

concept

formula

calculation $\times \div$

Sol

$$\vec{L} = \vec{r} \times \vec{p}$$

$$= (2\hat{i} + 3\hat{j}) \times (2\hat{i} + 5\hat{j})$$

$\Rightarrow \hat{i}$
 $\theta = 0$

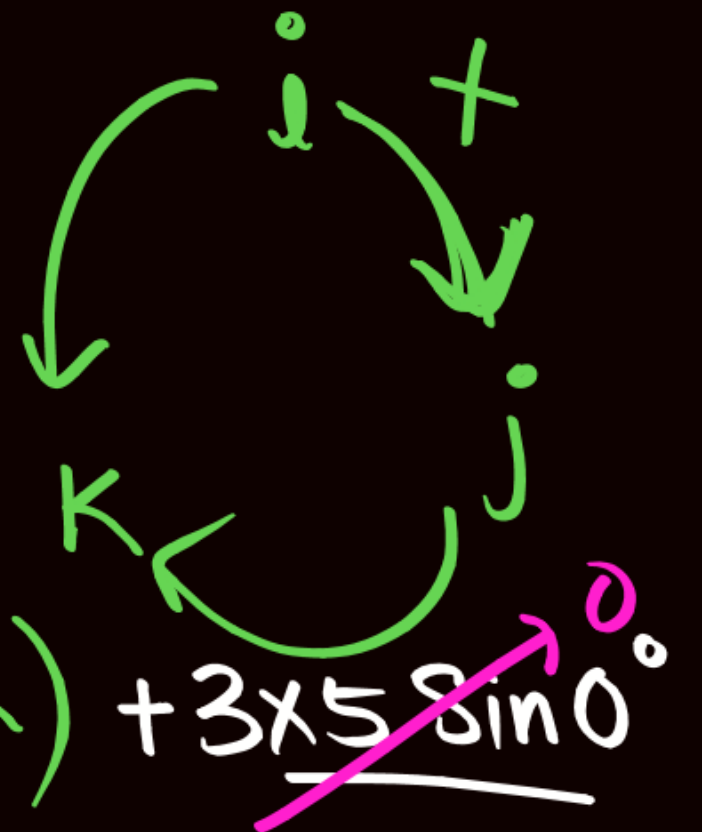
$$= \underline{2\hat{i}} \times \underline{2\hat{i}} + 2\hat{i} \times 5\hat{j} + 3\hat{j} \times 2\hat{i} + 3\hat{j} \times 5\hat{j}$$

$$= 2 \times 2 \cancel{\sin 0^\circ} + 2 \times 5 \sin 90^\circ \hat{k} + 3 \times 2 \sin 90^\circ (-\hat{k}) + 3 \times 5 \cancel{\sin 0^\circ}$$

$$= 10 \times 1 \hat{k} + 6 \times 1 \times (-\hat{k})$$

$$= 10\hat{k} + 6(-\hat{k})$$

τ



$$= 10\hat{k} + (-6\hat{k})$$

$$= 10\hat{k} - 6\hat{k}$$

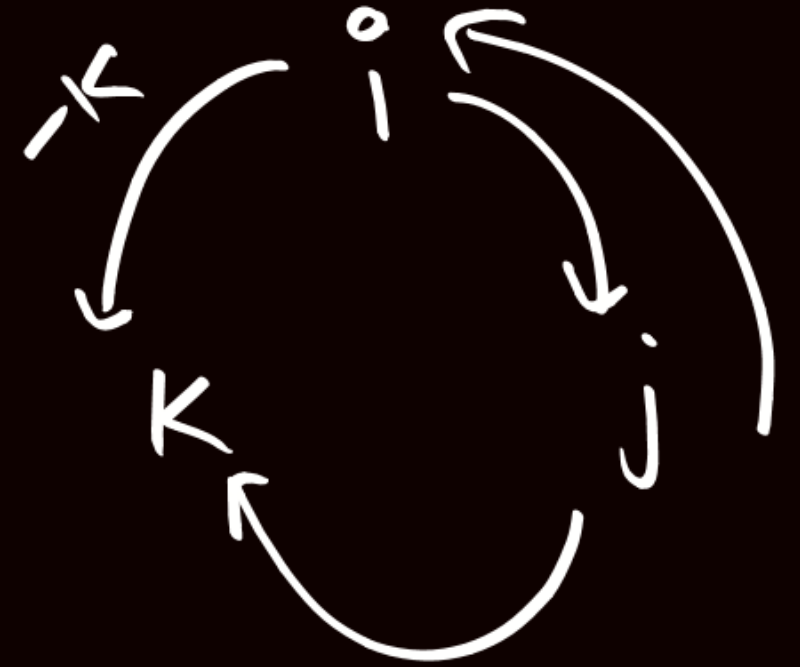
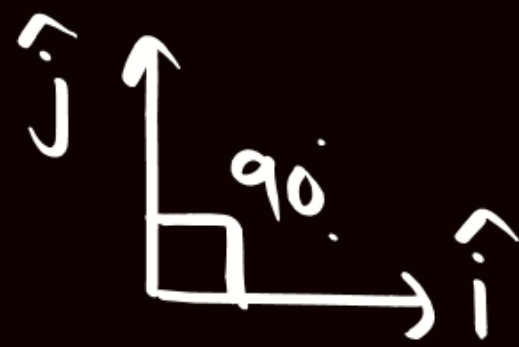
$$= \underline{\underline{4\hat{k} \text{ kg} \frac{\text{m}^2}{\text{s}}}}$$

$$\begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} \begin{array}{c} i \\ i \end{array} \quad \begin{array}{l} \theta = 0 \\ \sin 0 = 0 \end{array}$$

$$\begin{array}{c} j \\ j \end{array} \begin{array}{c} \uparrow \\ \uparrow \end{array} \quad \begin{array}{l} \theta = 0 \\ \sin 0 = 0 \end{array}$$

Numerical (2018-TKBOSE)

Q. $\vec{r} = \hat{i} + 2\hat{j}$
 $\vec{p} = 3\hat{i} + 4\hat{j}$ $\vec{L} = ?$



Sol $\vec{L} = \vec{r} \times \vec{p}$
 $= (\hat{i} + 2\hat{j}) \times (3\hat{i} + 4\hat{j})$

$$= \hat{i} \times 3\hat{i} + \hat{i} \times 4\hat{j} + 2\hat{j} \times 3\hat{i} + 2\hat{j} \times 4\hat{j}$$

$$= \cancel{1 \times 3 \sin 0^\circ} + 1 \times 4 \times \sin 90^\circ \hat{k} + 2 \times 3 \sin 90^\circ (-\hat{k}) + \cancel{2 \times 4 \sin 0^\circ}$$

$$= 4 \times 1 \hat{k} + 6 \times 1 \times -\hat{k}$$

$$= 4\hat{k} + -6\hat{k} = 4\hat{k} - 6\hat{k} = \underline{\underline{-2\hat{k}}}$$

Q $\vec{r} = 2\hat{k}$ $\vec{p} = 2\hat{k}$ $\vec{L} = ?$

Sol

$$\begin{aligned}\vec{L} &= \vec{r} \times \vec{p} \\ &= 2\hat{k} \times 2\hat{k} \\ &= 2 \times 2 \times \cancel{\sin 0^\circ} 0 \\ &= 4 \times 0 \\ &= 0 \text{ Kg m}^2/\text{s}\end{aligned}$$

YE ASAAN HAI [SUMMER SERIES-A 2026]

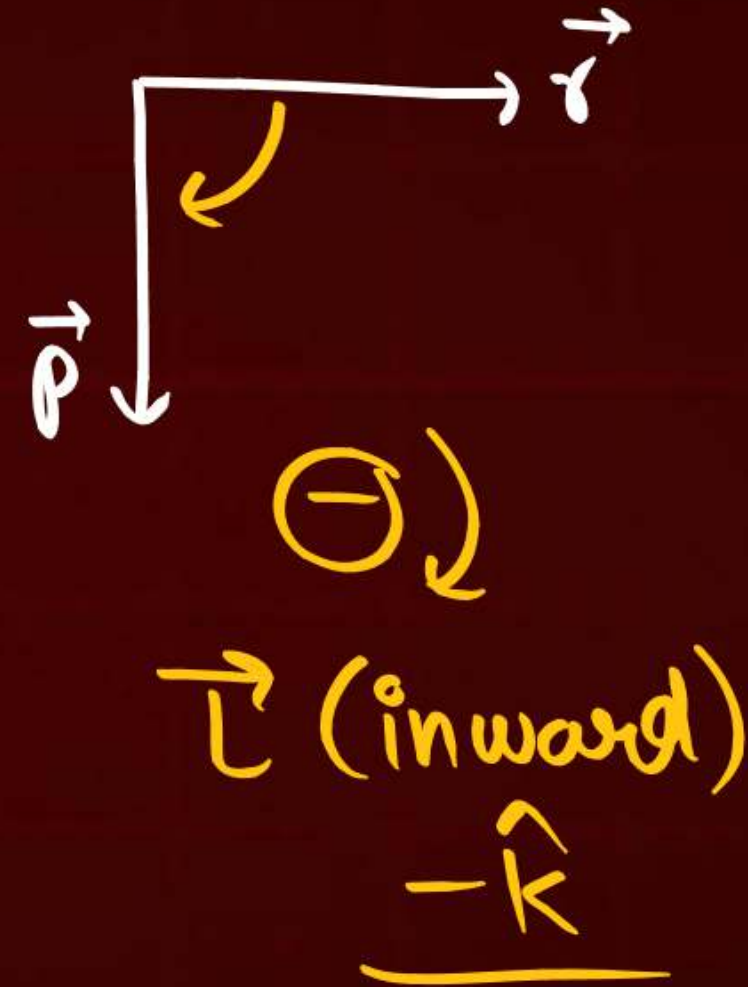
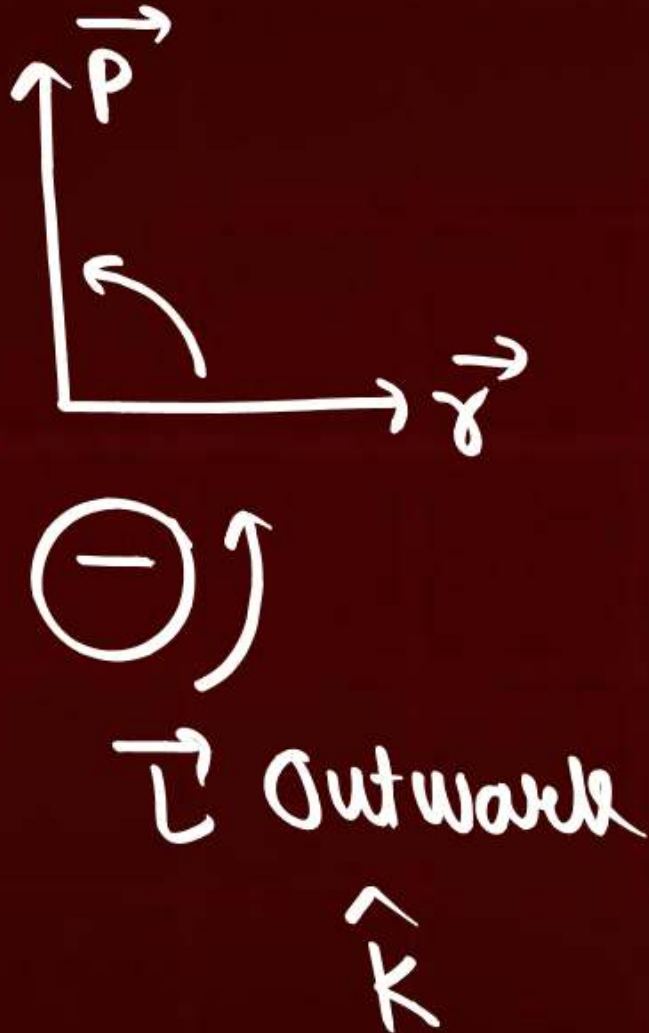
Unit of Angular Moment? JKBOSE

Sol //

$$\begin{aligned}
 \vec{L} &= \vec{r} \times \vec{p} \\
 &= \vec{r} \times m\vec{v} \\
 &= m \text{ kg m/s} \\
 &= \text{kg m}^2/\text{s}
 \end{aligned}$$

Direction of Angular Momentum

The direction of L is given by screw rule ✓



Jee-Advanced (2020)

Q $\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{L} = ?$

Board

$\vec{p} = 4\hat{i} + 5\hat{j} + 6\hat{k}$

Sol $\vec{L} = \vec{r} \times \vec{p}$

$= (2\hat{i} + 3\hat{j} + 4\hat{k}) \times (4\hat{i} + 5\hat{j} + 6\hat{k})$

$= \textcircled{3} \quad \textcircled{3} \quad \textcircled{3}$

$= \underline{9 \text{ terms}}$

Jee-Advanced. (2020)

Q $\vec{r} = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{p} = 4\hat{i} + 5\hat{j} + 6\hat{k}$

Sol

Determinant Method

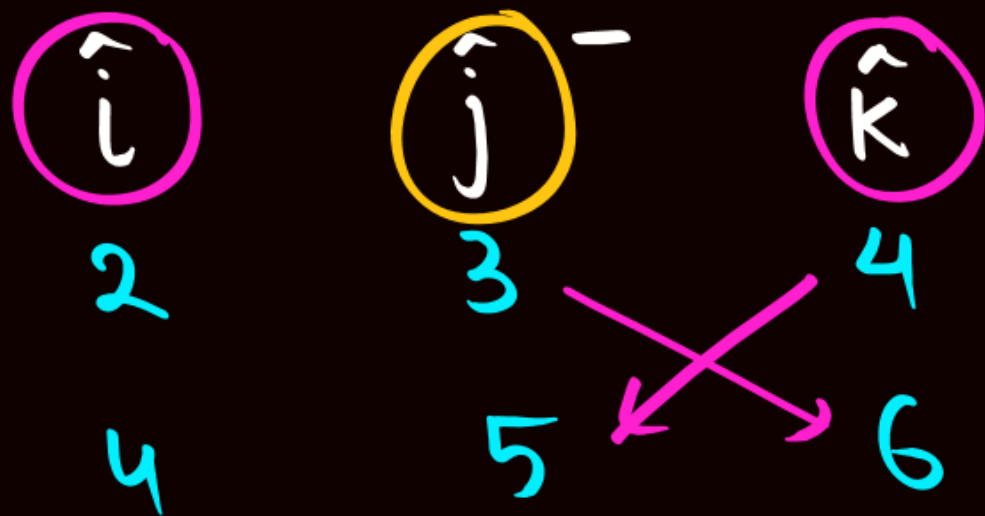
Trick

Answer - $\hat{i}(3 \times 6 - 5 \times 4) + -\hat{j}(2 \times 6 - 4 \times 4) + \hat{k}(2 \times 5 - 4 \times 3)$

$\hat{i}(18 - 20) - \hat{j}(12 - 16) + \hat{k}(10 - 12)$

$\hat{i} \times -2 - \hat{j} \times -4 + -2 \times \hat{k}$
 $-2\hat{i} + 4\hat{j} - 2\hat{k}$

== #



~~Topic~~

$$\vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

$$\vec{p} = 3\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\vec{r} = ?$$

Sol

$$\hat{i}$$

$$2$$

$$3$$

$$-\hat{j}$$

$$3$$

$$2$$

$$\hat{k}$$

$$5$$

$$4$$

Answer

$$\hat{i}(3 \times 4 - 2 \times 5) + -\hat{j}(2 \times 4 - 3 \times 5) + \hat{k}(2 \times 2 - 3 \times 3)$$

$$\hat{i}(12 - 10) - \hat{j}(8 - 15) + \hat{k}(4 - 9)$$

$$\hat{i}(2) - \hat{j}(-7) + \hat{k}(-5)$$

$$2\hat{i} + 7\hat{j} - 5\hat{k}$$

Torque

$\vec{\tau} \rightsquigarrow \underline{\text{Tau}}$

The Rotating effect of the force about an axis of rotation is called torque

OR

The Torque is a cross product of $\vec{r}$ and $\vec{F}$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$= r p \sin \theta (\hat{n}) \rightarrow \text{direction}$$

$\Downarrow$

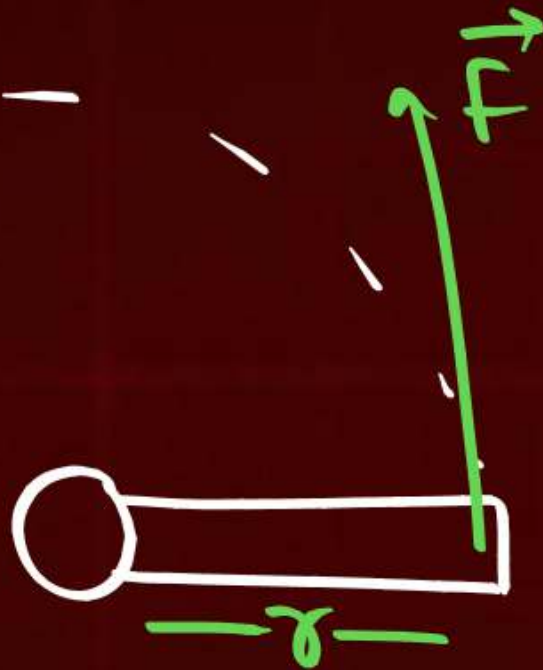
Screw Rule

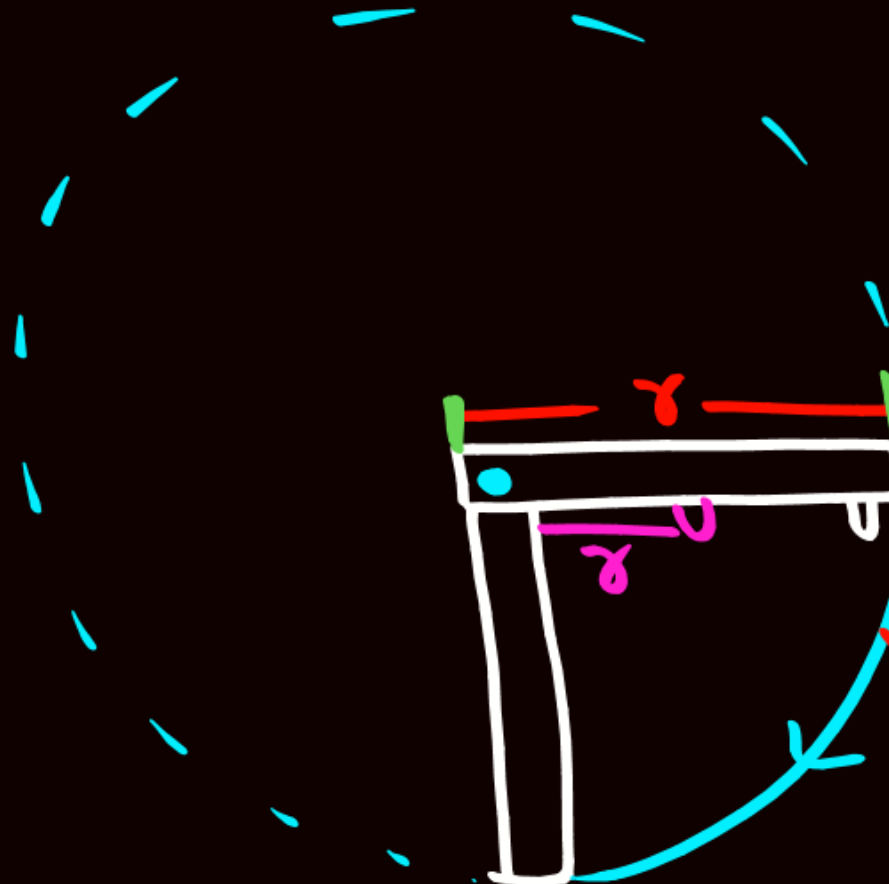
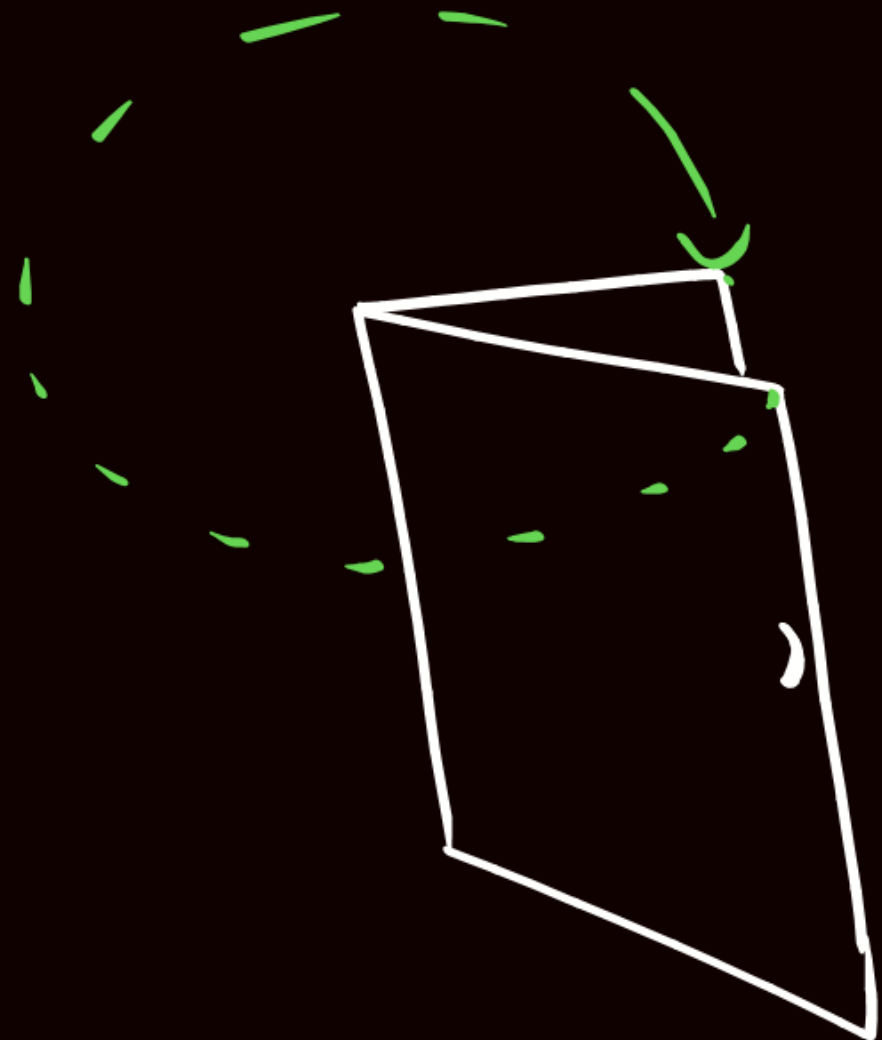
Unit

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$= \text{mN}$$

$$= \underline{\text{Nm}}$$



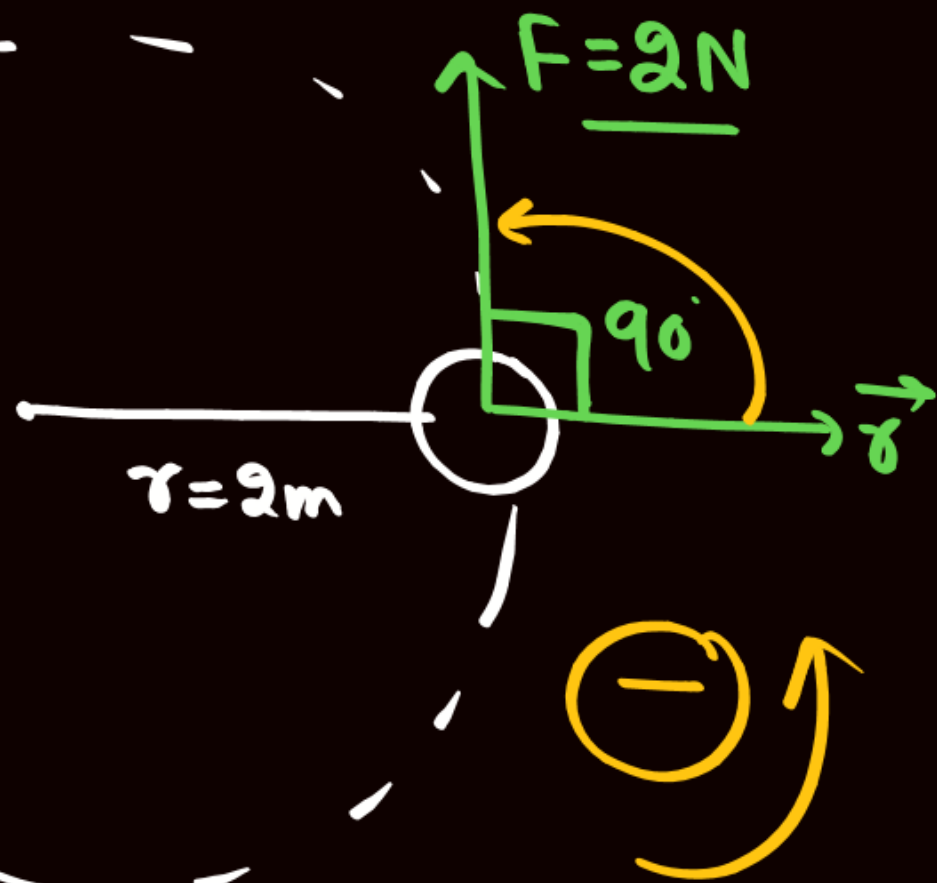


$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\frac{\vec{r}}{r} = \frac{\vec{F}}{F}$$

$$\frac{\vec{r}}{r} = \frac{\vec{F}}{F}$$

Q (numerical)



$$\boxed{\vec{\tau} = ?}$$

Sol

$$\vec{\tau} = \vec{r} \times \vec{p}$$

$$= r p \sin \theta \hat{n}$$

$$= 2 \times 2 \sin 90^\circ \hat{n}$$

$$= 4 \times 1 \hat{n} \text{ Direction}$$

$$= 4 \text{ outward}$$

$$= 4 \hat{k} \text{ Nm}$$

Direction of Torque

The direction of Torque is given by screw rule

Kar Ke Dikhao [NEET 2017]

A particle of mass 2 kg moves in a circle of radius 3 m with a velocity of 4 m/s. Find its angular momentum about the center.

“BELIEVE IN YOURSELF”

**“EVEN WHEN NO ONE
ELSE DOES.”**

Thank
You

